

VAPO: Value-Model-Based RL for Advanced Reasoning

60.4 on AIME 2024

Authors: ByteDance Seed
Base Model: Qwen2.5-32B
SFT Data: None

Why Value Models? Finer Credit Assignment for Long CoT

Value-Model-Free (GRPO, DAPO)

Uniform Advantage: Computes advantage based on final trajectory reward and assigns it uniformly across all tokens.

Limitation: Struggles with precise credit assignment when subtle step-level errors cause catastrophic failures.

Value-Model-Based (VAPO)

Token-Level Credit: Enables precise tracking of each action's impact on returns.

Variance Reduction: Provides lower-variance estimates per token.

Generalization: Well-trained value models generalize across samples.

Three Challenges Plaguing Value-Model-Based RL in Long-CoT

1

Value Model Bias

Bootstrapped learning over long trajectories compounds errors, leading to initial bias.

2

Heterogeneous Sequence Lengths

Variable lengths require different bias-variance tradeoffs. Uniform GAE fails to optimize both.

3

Reward Signal Sparsity

In long CoT tasks, only the final token receives a reward, exacerbating credit assignment.

VAPO Framework: Integrated Solutions

Value-Pretraining

Pre-trains value model to reduce initial bias

Decoupled-GAE

Separates GAE computation for robust value estimation

Length-Adaptive GAE

Dynamically adjusts λ in GAE based on response length

Clip-Higher

Asymmetric clipping to encourage exploration

Token-Level Loss

Normalizes loss at the token level

Positive Example LM Loss

Self-imitation learning on correct trajectories

Group-Sampling

Samples multiple responses for better baseline estimation

Length-Adaptive GAE: Dynamic λ Adjustment

Novel Technique Optimizing Bias-Variance Tradeoff for Variable Response Lengths

Short Responses

Lower λ

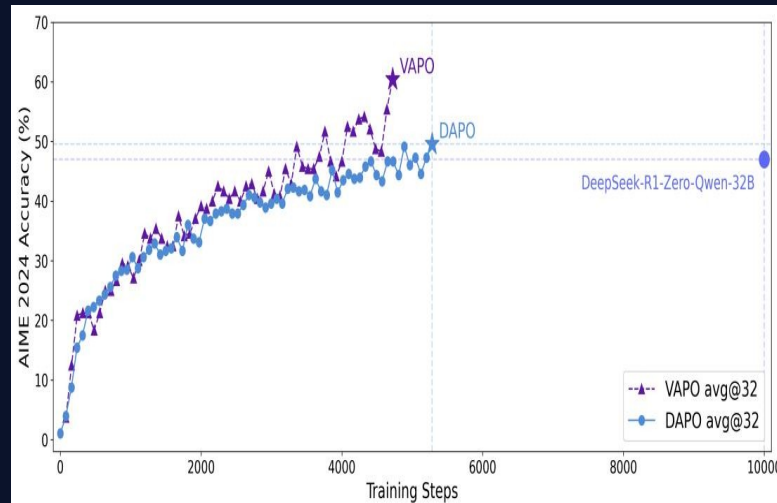
Prioritizes lower variance over bias.

Long Responses

Higher λ

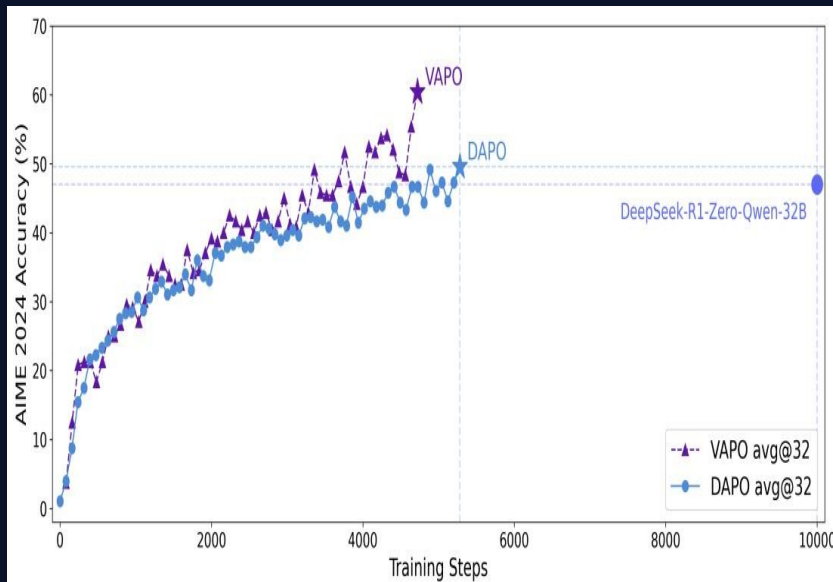
Prioritizes lower bias over variance, critical for long reasoning chains.

VAPO Scores 60.4 on AIME 2024



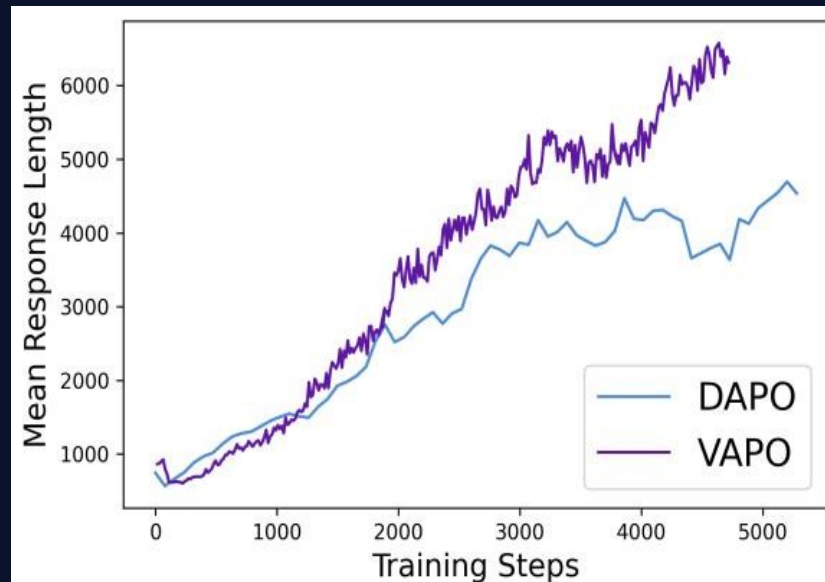
Training curve showing VAPO outperforming DAPO consistently

VAPO Reaches SOTA in 5K Steps



Training Accuracy Curve

Reaches 60% accuracy at step 4,500. DAPO plateaus near 50% at step 5,500.



Mean Response Length

Generates longer chains (~6,000 tokens) vs DAPO (~4,500), indicating deeper reasoning.

Ablation: Value-Pretraining Is Critical

Model Configuration	AIME24 avg@32	Delta
VAPO (Full)	60	-
w/o Group-Sampling	55	-5
w/o Positive Example LM Loss	54	-6
w/o Token-level Loss	53	-7
w/o Clip-Higher	46	-14
w/o Length-adaptive GAE	45	-15
w/o Decoupled-GAE	33	-27
w/o Value-Pretraining	11	-49

Key Takeaways: Value-Model-Based RL Is Back

- **Value-Model-Based Superiority:** Can surpass value-free methods when challenges (bias, heterogeneous lengths, reward sparsity) are systematically addressed.
- **Crucial Value Pretraining:** The single most important component (+49 points). Without it, value-model-based RL fails on long reasoning tasks.
- **Length-Adaptive GAE:** A novel and critical technique addressing the heterogeneous sequence length problem unique to long-CoT reasoning.
- **Exceptional Stability:** VAPO exhibits no training crashes across multiple runs, achieving SOTA efficiently (~5K steps).